

Project Log: Initiation

Project:	Smart Campus Network Optimization Platform
Log Date:	September 12, 2026
Log Type:	Initiation Log
Timeline:	September 15, 2026 - June 1, 2027

Overview

This log marks the formal start of the project's logging record. It captures the original project idea, its three core modules, and the planning work completed before day-to-day logging began. Going forward, each log will document what changed, was decided, or was achieved on a given date.

Project Concept

The project is a graduation project centered on building an optimization plan for a campus network, organized into three major parts: Wi-Fi optimization, network automation and monitoring, and multi-vendor network integration.

- **Wi-Fi Optimization:** Planning Wi-Fi coverage using heatmaps and relocating access points accordingly, paired with a custom-built access point featuring a 3D-designed case, dual antennas for signal strength, and a custom GUI built on OpenWRT.
- **Network Automation & Monitoring:** Python scripts that push Layer 2 configuration automatically based on input from a web server, alongside a monitoring script that analyzes the Layer 2 system and sends notifications/data to the same web server. Both run on a microcontroller (Raspberry Pi), with a physical button able to launch the servers and deploy the scripts.
- **Multi-Vendor Integration:** A connection between multi-vendor networks (e.g. Cisco and Huawei) using a custom-configured VPN tunnel built with GRE technology.

AI & Automation Extension

The original manual automation concept was extended with an AI layer that runs on the same Raspberry Pi hardware, without removing the manual automation path.

- The web server gains a chatbot interface: instead of buttons and fixed commands, the user can type a request (e.g. "make VLAN 10 on switch 1") in natural language.
- An AI system converts that text request into executable configuration code that can be pushed to a switch.
- n8n workflow automation ties the pipeline together: chatbot request → API call → code generation → a credentials step (device name, port number, target port, management IP, etc.) → a validator that checks the code for correctness and network safety → script execution against the network → confirmation message returned to the chat box.

- All coding components (scripts, Flask, the web server, and n8n) are intended to run on the Raspberry Pi.

Current Technical Setup

- Docker installed on Ubuntu 22.04, running n8n.
- n8n connected to an online AI API (Groq, compound-mini model).
- A secondary option under consideration: Flask 3.5 (lite), split into one component for code generation and another for credential handling/validation.

Team Structure & Timeline

The project is organized into three parallel teams running for the full project duration (Sep 15, 2026 - Jun 1, 2027), with monthly syncs and cross-team milestones to keep the modules aligned.

- Team 1 - Wi-Fi optimization and custom access point hardware.
- Team 2 - Automation, monitoring, AI pipeline, and the Raspberry Pi web server.
- Team 3 - Network topology and multi-vendor integration (Cisco lab devices, GRE tunnel design and testing).

Key Cross-Team Milestones

- Oct 15, 2026 - Proposal Freeze: scope, requirements, and lab access confirmed.
- Dec 15, 2026 - First-Cut Hardware: OpenWRT boots on the custom AP; GRE tunnel established.
- Jan 15, 2027 - Pi Becomes Central Host: automation, monitoring, web server, and controls unified on the Raspberry Pi.
- Feb 15, 2027 - Mid-Project Review: AP deployed on-site; AI pipeline wired end-to-end; topology extended.
- Apr 15, 2027 - Full System Integration: Wi-Fi, AI automation, and multi-vendor topology tested together.
- May 15, 2027 - Feature Freeze: testing and hardening only.
- Jun 1, 2027 - Final Delivery: working demo, documentation, and defense materials submitted.

Status

Project concept, technical approach, and team timeline are defined and documented. This log establishes the baseline against which future changes and decisions will be tracked.